

Irrationality after One Stabilization of Universally CH_0 -Trivial Cubic Threefolds

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Abstract

Does universal CH_0 -triviality force rationality after stabilization? We construct a non-isotrivial one-parameter family of smooth cubic threefolds X_b for which X_b is universally CH_0 -trivial but $X_b \times \mathbf{P}^1$ is irrational. More generally, we prove that $X \times \mathbf{P}^1$ is irrational for every smooth cubic threefold X .

The examples arise from the nonstandard A_5 -invariant cubic pencil. Its six elliptic axes have polarization matrix $6I - J$. Their oriented six-set determines a rank-ten symplectic envelope with five principal two-primary gluings; the golden orientation selects the Frobenius-conjugate exotic pair. The exotic slope is semisimple at two, while the three-primary gluing is scalar. A local mixed-adjugate argument therefore puts the primitive class $\Theta^4/4!$ of the intermediate Jacobian in the integral divisor-product lattice. Voisin's criterion gives universal CH_0 -triviality fibre by fibre.

For irrationality, we use the formal-monodromy atom of the big quantum connection. The cubic atom has primitive sixth-root eigenvalues. The projective-bundle formula gives two copies on $X \times \mathbf{P}^1$, whereas the atoms of every point, curve, or surface have only integral or half-integral formal exponents. Additivity and weak factorization prevent cancellation. Thus the cycle-theoretic and quantum detectors diverge on the same smooth fourfolds.

Keywords. cubic threefold; universal CH_0 -triviality; irrationality; intermediate Jacobian; quantum connection; weak factorization.

1 Introduction

A smooth projective variety is universally CH_0 -trivial if the degree map

$$\deg : CH_0(Y_L) \longrightarrow \mathbf{Z}$$

is an isomorphism for every field extension $L/\mathbf{C}$. This condition is a central consequence of a decomposition of the diagonal and obstructs stable rationality. Its vanishing, however, need not provide a rational parametrization. We ask how far the failure can persist after one explicit stabilization.

Our examples are fourfolds of the form $Y = X \times \mathbf{P}^1$, where X is a smooth cubic threefold in the nonstandard A_5 -invariant pencil. Two independent mechanisms act on the same Y . The intermediate Jacobian of X has an algebraic primitive minimal class; this gives universal CH_0 -triviality. The big quantum connection of every smooth cubic contains a formal-monodromy atom with primitive sixth roots; weak factorization shows that this atom survives the product with $\mathbf{P}^1$.

Theorem 1.1 (Separation after one stabilization). *Let $X \rightarrow B^\circ$ be the smooth locus of the nonstandard A_5 -invariant pencil of cubic threefolds. For every $b \in B^\circ(\mathbf{C})$,*

$$X_b \text{ is universally } CH_0\text{-trivial,} \quad X_b \times \mathbf{P}^1 \text{ is universally } CH_0\text{-trivial and irrational.}$$

The family is non-isotrivial.

The irrationality statement does not depend on the symmetry of the pencil.

Theorem 1.2 (One-step irrationality). *For every smooth complex cubic threefold X , the fourfold $X \times \mathbf{P}^1$ is irrational.*

We do not claim that X is stably irrational. From the second stabilization onward, weak-factorization centers can themselves have cubic dimension, and the low-dimensional exclusion used here no longer applies.

How the two proofs meet

The cycle-theoretic half starts from six elliptic quotients of the Fano surface. Their Rosati Gram matrix is $6I - J$, so any five axes give an isogeny $E^5 \rightarrow J(X)$ of polarization type $(1, 6, 6, 6, 6)$. The two-primary principal kernels form $\mathbf{P}^1(\mathbf{F}_4)$. Three retain the full simplex symmetry; the other two are exchanged by Frobenius and have stabilizer A_5 . The golden orientation reconstructed by the companion cubic selects one member of this exotic pair. At two its graph slope has minimal polynomial $t^2 + t + 1$; at three the gluing is scalar. Both local forms admit primitive mixed-adjugate formulas, which produce $\Theta^4/4!$ as an integral product of divisor classes.

The quantum half uses a different invariant. We count primitive sixth-root eigenvalues of formal monodromy on the atoms of the maximal big-quantum F -bundle. Cai's cubic calculation gives a nonzero count. The projective-bundle formula doubles it on $X \times \mathbf{P}^1$. If this fourfold were rational, weak factorization would express that count using centers of dimension at most two. Surface classification and the low-dimensional atom formula show that every such center has count zero.

Relation with the reconstruction series

The finite six-axis carrier first arose as the common output of distinct sparse-shadow reconstructions. The present paper is logically standalone: Section 2 defines its integral envelope and proves the geometric realization needed here. The series provenance explains why the exotic sheet comes with a preferred golden orientation; it is not a premise for the cycle or quantum calculations.

Proof map

Section 2 constructs the rank-ten symplectic envelope and identifies the exotic cubic gluing. Section 3 proves the local divisor-product theorem and applies it to the primitive minimal class. Section 4 proves Theorem 1.2 from formal monodromy and weak factorization. The final section combines the two detectors and records the exact scope boundary.

2 The integral envelope and the exotic cubic sheet

Let Ω be the six conjugate D_5 -subgroups of A_5 , equivalently the six Sylow-5 subgroups with their normalizers. Put

$$A(\Omega) = \ker(\mathbf{Z}^\Omega \xrightarrow{\Sigma} \mathbf{Z}), \quad A(\Omega)^\vee = \text{Hom}(A(\Omega), \mathbf{Z}).$$

The perfect evaluation pairing defines a unimodular alternating form on

$$\mathcal{H}_{\text{ref}}(\Omega) = A(\Omega)^\vee e \oplus A(\Omega)f, \quad \langle xe, yf \rangle = y(x).$$

This is the hyperbolic symplectic envelope of the oriented six-set.

Proposition 2.1 (Universal envelope). *The construction $\Omega \mapsto \mathcal{H}_{\text{ref}}(\Omega)$ is functorial for permutations of Ω . Rationally it is the unique rank-ten symplectic A_5 -module whose two Lagrangians are copies of the standard five-dimensional constituent and whose pairing restricts to the root-weight evaluation pairing.*

Proof. The permutation lattice and its augmentation kernel are functorial. The evaluation pairing between $A(\Omega)$ and its dual is perfect, so the displayed alternating form is unimodular. Over $\mathbf{Q}$, the augmentation module is absolutely irreducible and its invariant symmetric form is unique up to scale. Schur's lemma therefore writes every symplectic module with two such constituents as $A(\Omega)_{\mathbf{Q}} \otimes U$, where U is a symplectic plane. Requiring the two integral Lagrangians to pair perfectly fixes the root-weight lattice above. $\square$

The finite golden orientation enters at the prime two. Let

$$H_2 = \text{Aug}(\mathbf{F}_2^\Omega) / \langle \mathbf{1} \rangle.$$

Then H_2 is simple and $\text{End}_{A_5}(H_2) = \mathbf{F}_4$. A normalized conference operator B with $B^2 = 5I$ acts integrally on $\mathbf{Z}^\Omega$. Passing to the canonical saturation

$$D_6^\vee = \mathbf{Z}^\Omega + \mathbf{Z} \frac{\mathbf{1}}{2}, \quad \varphi = \frac{I + B}{2},$$

gives $\varphi^2 - \varphi - 1 = 0$. Its reduction on the commutator heart is one of the two primitive scalars $\omega, \omega^2 \in \mathbf{F}_4$, and replacing B by $-B$ applies Frobenius.

Proposition 2.2 (Golden selection of the exotic pair). *The two orientations of the conference class determine the two primitive scalars in $\text{End}_{A_5}(H_2) = \mathbf{F}_4$. Under orientation reversal,*

$$B \longmapsto -B, \quad \omega \longmapsto \omega^2.$$

Thus the golden-orientation torsor is canonically the Frobenius torsor of the exotic two-primary principal gluings.

Proof. Every column of $(I + B)/2$ has the same half-integral class modulo $\mathbf{Z}^\Omega$, so $D_6^\vee$ is the least stable over-lattice. The commutator submodule of $D_6^\vee/2D_6^\vee$ is the four-dimensional six-point heart and its quotient is a trivial $\mathbf{F}_4$ -line. The extension has no invariant $\mathbf{F}_4$ -line, so the restriction of $\bar{\varphi}$ to the heart is primitive. Finally $(I - B)/2 = 1 - \varphi$, and in characteristic two $1 - \omega = \omega^2$. $\square$

We now recall the geometric source of the same lattice. Let S be the Fano surface of a smooth member of the A_5 -pencil and $(J, \Theta) = \text{Alb}(S)$ its intermediate Jacobian. Each of the six D_5 -subgroups gives an elliptic quotient $q_H : J \rightarrow E_H$ and its Rosati dual $i_H : E_H \rightarrow J$.

Proposition 2.3 (Six-axis polarization). *After coherent identification of the elliptic factors, the Rosati Gram matrix of the six maps is*

$$G_6 = 6I_6 - J_6.$$

Any five axes therefore give an isogeny $E^5 \rightarrow J$ whose pulled-back polarization has Smith type $(1, 6, 6, 6, 6)$.

Proof. Roulleau's intersection table gives self-intersection zero for the six fibre classes and pair intersection 24. Two-transitivity makes the Rosati matrix have diagonal d and off-diagonal m . The missing trivial constituent gives $d + 5m = 0$. The minimal-class identity $[S] = \Theta^3/3!$, together with the polarized Riemann–Roch trace formula, turns the pair intersection into $d^2 - m^2 = 24$. The positive solution is $(d, m) = (5, -1)$, hence $G_6 = 6I - J$. A five-by-five minor has Smith form $(1, 6, 6, 6, 6)$. $\square$

Proposition 2.4 (Principal gluing packet). *The A_5 -stable two-primary principal kernels of the six-axis source are parametrized by*

$$\mathbf{P}^1(\mathbf{F}_4) = \mathbf{P}^1(\mathbf{F}_2) \sqcup \{\omega, \omega^2\}.$$

The three $\mathbf{F}_2$ -rational kernels are S_6 -stable. Each of the other two has stabilizer exactly A_5 , and the outer normalizer exchanges them. The intermediate Jacobian of the nonstandard A_5 -cubic component uses the latter pair. Its three-primary kernel is the scalar graph over the unique monodromy-invariant line in elliptic three-torsion.

Proof. At two the coefficient discriminant is the simple four-dimensional $\mathbf{F}_2 A_5$ -module H_2 , whose commutant is $\mathbf{F}_4$. The full discriminant is $H_2 \otimes_{\mathbf{F}_2} \mathbf{F}_2^2$. An A_5 -stable simple half is therefore a one-dimensional $\mathbf{F}_4$ -subspace of $\mathbf{F}_4^2$, and every such half is maximal isotropic for the tensor product of the coefficient form with the alternating multiplicity form. This gives $\mathbf{P}^1(\mathbf{F}_4)$.

The simplex group acts on the coefficient heart over $\mathbf{F}_2$. Hence the three lines defined over $\mathbf{F}_2$ retain S_6 . A primitive scalar in $\mathbf{F}_4$ breaks this action to the exceptional subgroup A_5 ; its two values are exchanged by the outer normalizer. Were the generic cubic kernel $\mathbf{F}_2$ -rational, its principal polarization would inherit the faithful S_6 -action. Strong Torelli would then give an S_6 -action on the generic cubic, contrary to the automorphism group of the nonstandard component [3, Section 5]. Thus the generic kernel is exotic. The kernel is a finite sub-local-system, so the same pair persists throughout the connected smooth locus.

At three the coefficient heart is simple with scalar commutant. The $\Gamma_0(3)$ multiplicity monodromy has one invariant line in elliptic three-torsion. Tensoring that line with the coefficient heart is therefore the unique A_5 - and monodromy-stable maximal isotropic half. $\square$

Proposition 2.2 labels the two exotic members by the finite golden orientation.

Hartlieb proves that its intermediate-Jacobian image is a one-dimensional special subvariety [3, Proposition 5.7]. Since the cubic period map is a locally closed embedding, the resulting smooth family is non-isotrivial. Only this landing and nonconstancy are used below; no identification of compactified parameter curves is needed for the separation theorem.

3 The primitive minimal class

Let (A, Θ) be a principally polarized abelian variety of dimension g . Write

$$\gamma_A = \frac{\Theta^{g-1}}{(g-1)!} \in H^{2g-2}(A, \mathbf{Z}).$$

The issue is not the rational algebraicity of γ_A , but its membership in the integral lattice generated by ordinary products of divisor classes.

We first isolate the normalization that will be used throughout. If $C_1, \dots, C_d$ are symmetric endomorphisms of a free module, write

$$\text{Mdet}(C_1, \dots, C_d) = [t_1 \cdots t_d] \det\left(\sum t_i C_i\right)$$

and define $\text{Madj}(C_1, \dots, C_{d-1})$ in the same way from the adjugate. Thus these are coefficient operators, not derivatives divided by a factorial.

Lemma 3.1 (Primitive mixed coefficients). *On $\text{Sym}_d(\mathbf{Z})$, the mixed determinant has image $\mathbf{Z}$, and the entries of the mixed adjugates span $\text{Sym}_d(\mathbf{Z})$. More precisely,*

$$\text{Mdet}(E_{11}, \dots, E_{dd}) = 1,$$

while the coefficient of $\prod_{r \neq i} t_r$ in $\text{adj}(\sum_{r \neq i} t_r E_{rr})$ is E_{ii} , and inserting $E_{ij} + E_{ji}$ together with the E_{rr} , $r \neq i, j$, gives $-E_{ij} - E_{ji}$.

Proof. The first two assertions are the diagonal monomials in the Leibniz formulas for the determinant and its cofactors. For the last assertion the only surviving permutations use the transposition (ij) , whose sign is minus. Every displayed coefficient is therefore ± 1 . In particular neither a factorial nor the usual factor two for an off-diagonal quadratic term occurs. $\square$

Theorem 3.2 (Semisimple graph slopes). *Let p be a prime and let $T \in \text{Sym}_g(\mathbf{F}_p)$. Form the principally polarized quotient of the elliptic-power source with coefficient polarization pI_g by the maximal isotropic graph of T . If the minimal polynomial of T is squarefree, then*

$$\gamma_A \in \text{im}(\text{Sym}^{g-1} \text{NS}(A) \longrightarrow H^{2g-2}(A, \mathbf{Z})).$$

Proof. Work over $\mathbf{Z}_p$, and pass to a finite unramified extension over which the squarefree minimal polynomial splits. Self-adjoint eigenspaces for distinct roots are orthogonal and nondegenerate. They lift, without dividing by two, to an orthogonal decomposition with blocks pB_λ , where each B_λ is symmetric and unimodular.

On a scalar slope block, the integral divisor lattice contains $p\text{Sym}(B_\lambda)$. Fill every non-target block by its primitive mixed determinant from Lemma 3.1. On the target block, its degree-one-lower mixed adjugates give every diagonal matrix unit and every symmetric off-diagonal matrix unit with coefficient ± 1 . Because the blocks have disjoint support, taking one coefficient from each block simply multiplies these primitive coefficients; no multinomial factor appears. Summing over target blocks constructs the cofactor of $\perp_\lambda pB_\lambda$, which is the pullback of γ_A . No factorial or off-diagonal factor two is introduced. Membership descends from the unramified extension by faithful flatness. Away from p , the quotient isogeny identifies the integral divisor and cohomology lattices, so local membership globalizes. $\square$

The scalar version of the same mixed-adjugate construction applies to the three-primary block of the six-axis polarization. We can therefore use the two local forms isolated in Section 2.

Lemma 3.3 (Local chart for the six-axis quotient). *Let $G = 6I_5 - J_5$. At each prime $p \mid 6$, the coefficient lattice has an orthogonal Jordan decomposition*

$$(\mathbf{Z}_p^5, G) \simeq U_0 \perp pU_1,$$

where U_0 has rank one and is unimodular and U_1 has rank four and is unimodular. The principal quotient is trivial on U_0 . On U_1/pU_1 , its graph slope has minimal polynomial $t^2 + t + 1$ at $p = 2$, while at $p = 3$ it is scalar on the simple coefficient constituent.

Proof. The Smith form of G is $(1, 6, 6, 6, 6)$, but here the orthogonal chart can also be written down. Let $e_1, \dots, e_5$ be the standard basis. Since $\langle e_1, e_1 \rangle = 5$ and $\langle e_1, e_j \rangle = -1$, the vectors

$$v_j = e_j + \frac{1}{5}e_1 \quad (2 \leq j \leq 5)$$

span the orthogonal complement of e_1 over $\mathbf{Z}_p$. In this basis its Gram matrix is

$$\frac{6}{5}(5I_4 - J_4).$$

For $p = 2, 3$, both 5 and $\det(5I_4 - J_4) = 5^3$ are units. We have therefore exhibited a unimodular line orthogonal to p times a unimodular rank-four block. The unit block has no discriminant and hence no gluing choice.

At two the discriminant heart is the six-point module H_2 , and Proposition 2.2 identifies the selected endomorphism with $\omega \in \mathbf{F}_4$; its polynomial is $t^2 + t + 1$. At three the coefficient heart is simple with scalar commutant, and the unique monodromy-invariant elliptic three-torsion line makes the graph scalar. $\square$

Theorem 3.4 (Minimal class for the exotic cubic family). *For every smooth member X_b of the nonstandard A_5 -cubic pencil, with intermediate Jacobian (J_b, Θ_b) ,*

$$\frac{\Theta_b^4}{4!} \in \text{im}(\text{Sym}^4 \text{NS}(J_b) \longrightarrow H^8(J_b, \mathbf{Z})).$$

Proof. The five-axis source has polarization type $(1, 6, 6, 6, 6)$, so only the primes two and three can contribute an integral defect. At two the exotic graph slope acts on the simple heart through $\omega \in \mathbf{F}_4$; its minimal polynomial $t^2 + t + 1$ is squarefree. Theorem 3.2 therefore gives the primitive local class. At three the monodromy line makes the gluing scalar on the simple coefficient block, and the scalar mixed-adjugate formula gives the same conclusion. Away from two and three the source and quotient lattices are unimodular. Local membership at every prime is equivalent to integral membership in the divisor-product lattice.

Lemma 3.3 identifies the actual geometric lattice, not merely an abstract gluing with the same Smith form. The graph kernel is a finite sub-local-system on each connected smooth component of the pencil, so the same local presentation holds on every fibre. At a fibre with additional Néron–Severi classes the divisor-product image can only increase. $\square$

Voisin identifies this primitive class as the exact obstruction relevant to cubic threefolds.

Corollary 3.5 (Universal CH_0 -triviality). *Every smooth member of the nonstandard A_5 -cubic pencil is universally CH_0 -trivial.*

Proof. By Theorem 3.4, the primitive minimal class of the intermediate Jacobian is algebraic. Voisin’s Corollary 4.4 [6] says that, for a smooth complex cubic threefold, this is equivalent to universal CH_0 -triviality. $\square$

The argument is fibrewise. It constructs neither a horizontal minimal cycle over the pencil nor a relative decomposition of the diagonal. Neither is needed for Corollary 3.5.

4 The primitive-sixth-root atom

We use the geometric atomic F -bundle invariant of Katzarkov–Kontsevich–Pantev–Yu [4]. Only three features enter: its values lie in the free abelian group on atom-equivalence classes, it is additive under blowups, and a rank- r projective bundle contributes r copies of the base.

Definition 4.1. For a geometric atom α , let $\text{FM}(\alpha)$ be the multiset of formal-monodromy eigenvalues on its Levelt–Turrittin regular-singular factors, after the exponential parts have been separated. Define

$$\nu_6(\alpha) = \text{mult}_{e^{\pi i/3}} \text{FM}(\alpha) + \text{mult}_{e^{-\pi i/3}} \text{FM}(\alpha).$$

Extend ν_6 additively to the free atom group.

The definition uses eigenvalues in $\mathbf{C}^\times$, not chosen logarithms. Integral powers of the loop coordinate shift residue exponents by integers and leave ν_6 unchanged.

Lemma 4.2 (Formal isomonodromy). *Along a connected component of the reduced unramified spectral cover, the formal-monodromy eigenvalue multiset of an atom is constant.*

Proof. After finite formal ramification and separation of exponential factors, one block has the form

$$\nabla = d - d\Phi(s, u) + R(s) \frac{du}{u} + \sum_i \Gamma_i(s, u) ds_i,$$

where Γ_i is formal-regular after the exponential term is removed. The coefficient of $du/u \wedge ds_i$ in $\nabla^2 = 0$ is

$$\partial_{s_i} R = [\Gamma_i(s, 0), R].$$

Hence the characteristic polynomial of R , and therefore that of $\exp(2\pi i R)$, is locally constant. Descent only permutes the separated blocks. $\square$

Proposition 4.3 (Cubic and low-dimensional atoms). *Let X be a smooth cubic threefold.*

- (i) *The cubic atom α_X satisfies $\nu_6(\alpha_X) > 0$.*
- (ii) *Every atom represented by a point, a curve, or a smooth projective surface has $\nu_6 = 0$.*

Proof. Cai’s Proposition 6 gives formal residue classes $\pm 1/6$ in the rank-two zero-eigenvalue block of the big cubic quantum connection [2]. The same block is the unsplit cubic zero atom in the maximal big-quantum F -bundle of [4, Example 6.21]; scalar extension does not change its formal monodromy. This proves (i).

For a minimal surface with nef canonical class, the parity-corrected connection has nilpotent residue by [4, Claim 6.15]. Undoing the parity correction permits only eigenvalues 1 and -1 . Ruled surfaces reduce by the projective-bundle formula to curves, and nonminimal surfaces reduce by point blowups to their minimal models. Points and curves likewise have only integral or half-integral residues. The classification of smooth projective surfaces exhausts the possibilities and proves (ii). $\square$

Proof of Theorem 1.2. Let $Y = X \times \mathbf{P}^1$. The projective-bundle formula gives

$$\text{CFF}(Y) = 2 \text{CFF}(X),$$

so Proposition 4.3 gives $\nu_6(Y) > 0$.

Suppose that Y were rational. Weak factorization of a birational map $Y \dashrightarrow \mathbf{P}^4$ gives, in the free abelian atom group,

$$\text{CFF}(Y) + \sum_i (r_i - 1) \text{CFF}(Z_i) = \text{CFF}(\mathbf{P}^4) + \sum_j (s_j - 1) \text{CFF}(W_j),$$

where every smooth center Z_i, W_j has dimension at most two. Apply ν_6 . Proposition 4.3(ii) kills every center term, and the projective-bundle formula over a point gives $\nu_6(\mathbf{P}^4) = 0$. The left-hand side retains $\nu_6(Y) > 0$, a contradiction. $\square$

The proof uses the free atom group, not a signed informal block ledger. In particular the two copies supplied by $\mathbf{P}^1$ are positive multiplicities and cannot cancel one another.

5 The separation theorem

Proof of Theorem 1.1. Corollary 3.5 gives universal CH_0 -triviality of X_b . The projective bundle formula for Chow groups gives the same property for $X_b \times \mathbf{P}^1$. Theorem 1.2 proves that this fourfold is irrational. Non-isotriviality follows from the nonconstant intermediate-Jacobian period map on the A_5 -component. $\square$

The theorem separates two tests on one variety. The algebraic class $\Theta^4/4!$ removes the decomposition-of-diagonal obstruction. The primitive-sixth-root atom survives one stabilization and obstructs a rational factorization. The first mechanism is special to the exotic A_5 -family; the second holds for every cubic threefold.

The result stops after one stabilization. For $X \times \mathbf{P}^m$ with $m \geq 2$, weak-factorization centers may have dimension at least three and can carry cubic-type atoms. The low-dimensional exclusion in Proposition 4.3 no longer proves noncancellation. Likewise, the fibrewise minimal-class argument gives no relative universal cycle over the pencil. These are separate problems and are not implicit in the theorem above.

The reconstruction series ends with a finite marked carrier. Its maximal golden normalization supplies the $\mathbf{F}_4$ -sheet used here, and the six elliptic axes realize the corresponding integral symplectic envelope. The geometric consequence is therefore more than a second occurrence of the same A_5 -set: the recovered orientation selects the principal gluing on which the primitive minimal class becomes integral. On the resulting fourfolds, universal CH_0 -triviality and rationality part company.

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